

CÓDIGO: <i>Code</i>	798519	DESIGNAÇÃO: <i>Name</i>	LIT.KETOSTERIL COM REV MY	ELABORADO POR: <i>Made by</i>	José Duarte
VERSÃO: <i>Version</i>	/00	PROVA: <i>Proof</i>	02	DATA: <i>Date</i>	17 Abr.2015
FICHA TÉCNICA: <i>Technical Sheet</i>	798519	ESCALA: <i>Scale</i>	1:1		

CORES: Colours	● Black ● <i>Cut</i> ○ ○ ○
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INSTRUCTIONS FOR USE - READ CAREFULLY!
Fresenius Kabi Deutschland GmbH
D-61346 Bad Homburg v.d.H.

Ketosteril tablets

Film-coated tablets

Composition	Calcium-3-methyl-2-oxo-valerate	67 mg
One film-coated tablet contains:	(α -Ketoanalogue to Isoleucine, Calcium salt)	
	Calcium-4-methyl-2-oxo-valerate	101 mg
	(α -Ketoanalogue to Leucine, Calcium salt)	
	Calcium-2-oxo-3-phenyl- propionate	68 mg
	(α -Ketoanalogue to Phenylalanine, Calcium salt)	
	Calcium-3-methyl-2-oxo-butyrate	86 mg
	(α -Ketoanalogue to Valine, Calcium salt)	
	Calcium-DL-2-hydroxy-4-(methyl- thio) butyrate	59 mg
	(α -Hydroxyanalogue to Methionine, Calcium salt)	
	L-Lysine-acetate	
	($\approx$ 75 mg L-Lysine)	105 mg
	L-Threonine	53 mg
	L-Tryptophan	23 mg
	L-Histidine	38 mg
	L-Tyrosine	30 mg
	Total nitrogen content/tablet	36 mg
	Calcium/tablet	1.25 mmol 50mg

Product description: Oblong, yellow film-coated tablet

Indications Prevention and therapy of damages due to faulty or deficient protein metabolism in chronic renal insufficiency in connection with limited protein in food of 40 g per day (for adults) and less; i.e. generally in patients with a glomerular filtration rate (GFR) between 5 and about 15 **ml/minutes**.

Application and Dosage For oral use.
If not otherwise prescribed, take 4 to 8 tablets three times a day during meals. Swallow whole.
This dosage applies to adults (70 kg/BW).

Contraindications -Hypersensitivity to the active substances or to any of the

excipients
-Hypercalcaemia
-Disturbed amino acid metabolism

Side effects Hypercalcaemia may develop. In this case it is recommended to decrease vitamin D intake. If the hypercalcaemia persists, reduce the dosage of Ketosteril as well as other source of calcium. The serum calcium level should be monitored regularly. Ensure sufficient calorie intake. No experience has been gained so far with the administration in paediatric patients.

Special warnings and precautions for use In the presence of hereditary phenylketonuria, attention should be given to the fact that Ketosteril® contains phenylalanine.

Interactions with other drugs: Concomitant administration of calcium-containing drugs may cause or aggravate elevated serum calcium levels. Uraemic symptoms improve under therapy with Ketosteril®. Thus, in case of aluminium hydroxide administration, the dose of this drug has to be reduced if necessary. Serum phosphate levels should be monitored for a decrease.Drugs that form sparingly soluble compounds with calcium (e.g. tetracyclines, quinolones such as ciprofloxacin and norfloxacin as well as drugs containing iron, fluoride or estramustine) should not be taken at the same time with Ketosteril® to avoid disturbed absorption of the active substances. An interval of at least two hours should elapse between the ingestion of Ketosteril® and these drugs. The susceptibility to cardioactive glycosides, and hence the risk for arrhythmia will increase if Ketosteril® produces elevated serum calcium levels). Monitoring of the serum phosphate levels is needed in case of concomitant administration of aluminium hydroxide.

Pregnancy and lactation There are no adequate data from the use of Ketosteril® in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development. Caution should be exercised when prescribing to pregnant women. No experience has been made so far with the use during lactation. No case of overdose has been reported.

Overdose Ketosteril® tablets are administered for nutrition therapy in chronic kidney disease.

Pharmacodynamic properties Ketosteril® allows the intake of essential amino acids while minimising the amino-nitrogen intake. Following absorption, the keto- and hydroxy-analogues are transaminated to the corresponding essential amino acids by taking nitrogen from non-essential amino acids, thereby decreasing the formation of urea by re-using the amino group. Hence, the accumulation of uraemic toxins is reduced. Keto and hydroxy acids do not induce hyperfiltration of the residual nephrons. Ketoacid containing supplements exert a positive effect on renal hyper phos-phataemia and secondary hyperparathyroidism. Moreover, renal osteodystrophy may be improved. The use of Ketosteril® in combination with a very low protein diet allows to reduce nitrogen intake while preventing the deleterious consequences of inadequate dietary protein intake and mainutrition.

Pharmacokinetic properties The plasma kinetics of amino acids and their integration in the metabolic pathways are well established. It should nevertheless be noted that in uraemic patients, the cause of the changed plasma levels, which occur frequently in these patients, does not seem to be the absorption of the supplied amino acids, i. e. the absorption itself is not disturbed.The changed plasma levels seem to be due to impaired post-absorptive kinetics, which can be detected in a very early stage of the disease. In healthy individuals, the plasma levels of ketoacids increase within 10 min after oral administration. Increases of up to the 5-fold the baseline levels are achieved. Peak levels occur within 20-60 min, and after 90 min levels stabilise in the range of the base levels. Gastrointestinal absorption is thus very rapid. The simultaneous increases in the levels of the ketoacids and the corresponding amino acids show that the ketoacids are transaminated very rapidly. Due to the physiological utilisation pathways of ketoacids in the body it is likely that exogenously supplied ketoacids are very rapidly integrated into the metabolic cycles. Ketoacids follow the same catabolic pathways as classical amino acids. No specific study on ketoacid excretion has been performed to date.

Duration of Application Ketosteril tablets are given as long as the glomerular filtration rate

(GFR) is between 5 and about 15mL/minute. Simultaneously food should contain 40g/day protein or less (adults).

Presentation Pack containing 20 and 100 film-coated tablets.

Shelf life 3 years

Storage Do not store above 30 °C
Store in the original package and keep the blisters tightly closed to protect contents from moisture
Do not use Ketosteril tablets after expiry date!
Keep out of the reach of children.

Manufactured by Labesfal - Laboratórios Almiro, S.A.
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Portugal

For Fresenius Kabi Deutschland GmbH
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Date of revision: August 2014